

MATH 142: Final Exam
Spring —₁ 2026
04/30/2026
150 Minutes

Name: _____

Write your name on the appropriate line on the exam cover sheet. This exam contains 16 pages (including this cover page) and 15 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	
11	10	
12	10	
13	10	
14	10	
15	10	
Total:	150	

1. Showing all your work, complete the following:

(a) (5 points) Evaluate $\int \cos^3 \theta \, d\theta$

(b) (5 points) Set-up but *do not* evaluate an integral which computes the arclength of $y = \frac{2}{7} x^{7/2}$ on $[0, 1]$.

2. (10 points) Evaluate the integral below.

$$\int \frac{1}{(\sqrt{16-x^2})^3} dx$$

3. (10 points) Compute the following integral:

$$\int \frac{13x - 8}{(x - 2)(x + 1)} dx$$

4. (10 points) Showing all your work, compute the integral below.

$$\int_0^{\pi} x \sin(7x) \, dx$$

5. (10 points) Determine whether the following improper integral converges or diverges. Be sure to fully justify your response.

$$\int_0^{\infty} \frac{dx}{(x+4)^2}$$

6. Determine whether the following series converge or diverge. Be sure to fully justify your response.

(a) (5 points) $\sum_{n=1}^{\infty} \frac{3n+5}{7n-4}$

(b) (5 points) $\sum_{n=0}^{\infty} \left(\frac{1}{2^n} - \frac{1}{2^{n+1}} \right)$

$$\begin{aligned} \sum_{n=0}^{\infty} \left(\frac{1}{2^n} - \frac{1}{2^{n+1}} \right) &= \lim_{N \rightarrow \infty} \sum_{n=0}^N \left(\frac{1}{2^n} - \frac{1}{2^{n+1}} \right) \\ &= \lim_{N \rightarrow \infty} \left[\left(\frac{1}{2^0} - \frac{1}{2^1} \right) + \left(\frac{1}{2^1} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{2^3} \right) + \cdots + \left(\frac{1}{2^N} - \frac{1}{2^{N+1}} \right) \right] \\ &= \lim_{N \rightarrow \infty} \left[1 - \frac{1}{2^{N+1}} \right] \\ &= 1 \end{aligned}$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{2^n} - \frac{1}{2^{n+1}} \right) = \sum_{n=0}^{\infty} \left(\frac{2}{2^{n+1}} - \frac{1}{2^{n+1}} \right) = \sum_{n=0}^{\infty} \left(\frac{2-1}{2^{n+1}} \right) = \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} \right) = \sum_{n=0}^{\infty} \frac{1}{2} \left(\frac{1}{2} \right)^n$$

7. (10 points) Determine whether the following series converges or diverges. Be sure to fully justify your response. If the series converges, find its sum.

$$\sum_{n=2}^{\infty} 3 \left(-\frac{2}{5} \right)^n$$

8. (10 points) Being sure to fully support your reasoning, determine whether the series given below converges or diverges.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{7+2n}$$

9. Determine whether the following series converge or diverge. Be sure to fully justify your response.

(a) (5 points) $\sum_{n=0}^{\infty} \frac{(-3)^n}{n!}$

(b) (5 points) $\sum_{n=1}^{\infty} \left(\frac{3n+4}{2n} \right)^n$

10. (10 points) Determine whether the series below diverges, converges conditionally, or converges absolutely. Include all necessary justifications. [Hint: $n > \ln(n+1)$ for $n \geq 1$.]

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n+1)}$$

11. Determine the center, radius, and interval of convergence for the following power series:

(a) (5 points) $\sum_{n=0}^{\infty} n! (x + 5)^n$

(b) (5 points) $\sum_{n=0}^{\infty} \frac{x^n}{e^{n^2}}$ [Hint: Use the Root Test]

12. (10 points) Find the center, radius, and interval of convergence for the following power series:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n \sqrt{n}} (x-1)^n$$

13. (10 points) Find the third-degree Taylor polynomial, $T_3(x)$, for $f(x) = \sqrt{x}$ centered at $x = 1$.

14. Use a known Maclaurin series to find a...

(a) (5 points) ... power series representation for $\frac{1}{1+x^2}$.

(b) (5 points) ... series approximation to $\int_0^1 \ln(1+x^2) dx$.

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15. (10 points) Determine the sum of the following series:

(a) $12 - 6 + 3 - \frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \cdots$

(b) $\sum_{n=0}^{\infty} (-1)^n \frac{(0.3)^{2n+1}}{(2n+1)!}$

(c) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$